GRADE 1 · INTRO DEBUGGING

What Is a Bug?

A bug makes the code do the wrong thing

Name: .... Date: ....

★ **I can run a program and tell if it has a bug.**

What is a bug?

A bug is a mistake in the code that makes the wrong thing happen. To find a bug, you run the program and check: did Bit reach the target? If not, there is a bug. Bit starts on the number in the start block. The target is 4.

Bit's number path (target is 4)

0	1	2	3	4 target	5

Program A

start at 1

forward 3

1 Run Program A

Run Program A from 1. What number does Bit land on? Did it reach the target (4)? Is there a bug?

____

____

Program B

start at 1

forward 1

2 Run Program B

Run Program B from 1. What number does Bit land on? Did it reach the target (4)? Is there a bug?

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3 You try — which one?

Which program has a bug, A or B? In your own words, what is a bug?

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Big idea

A bug is code that does the wrong thing. You find a bug by running the program and checking the result against what you wanted. In the next sheets you will FIX bugs.

TEACHER GUIDE · GRADE 1 · INTRO DEBUGGING

What Is a Bug?

Quick facilitation notes & answer key

What this worksheet teaches

Students learn that a "bug" is an instruction set that gives the wrong result, and that you find one by running the program and comparing where Bit lands to the target. They run a correct program and a buggy one and decide which has the bug. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: tape a number line 0 to 5 on the floor and mark the target, so students can walk the buggy program and feel Bit miss.

How to lead it (about 20 minutes)

1. Show them (you do). Run Program A: it lands on the target — no bug. Then say a bug is when the program lands on the wrong number.
2. Do one together. Exercise 1 — Program A reaches 4, so there is no bug.
3. Let them try. Students run Program B (Ex 2), then decide which program has the bug and say what a bug is (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Program A: 1, forward 3 → 4. It reaches the target, so there is NO bug.

Exercise 2 — Program B: 1, forward 1 → 2. It does not reach 4, so there IS a bug.

Exercise 3 — Program B has the bug. A bug is a mistake in the program that makes it do the wrong thing (here Bit lands on 2 instead of 4).

If students get stuck — and ways to adjust

Watch for: deciding there is a bug before running the program. Always run it and compare to the target first.

Make it easier: walk both programs on the floor line.

Make it harder: ask them to change Program B so it also reaches 4.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a program, change a block to fix it, and explain what the change did.

You'll know it worked when

The child runs both programs, names Program B as the buggy one, and explains what a bug is.